

Vulnerability Assessment Report for a Live Website

Cyber Security Task 1 – Future Interns

Name: RUCHITH PATHANI

Internship Name: Cyber Security Internship

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Assessment Type: Passive Vulnerability Assessment

Tools Used: Nmap, OWASP ZAP, Browser DevTools, Kali Linux

INTRODUCTION

A Vulnerability Assessment is the process of identifying, analyzing, and reporting security weaknesses in a website or web application. The purpose of this assessment is to evaluate the security posture of the target website using security assessment tools such as Nmap, OWASP ZAP, and Browser Developer Tools.

Objectives:

- Identify security vulnerabilities in the website
- Analyze risks associated with discovered vulnerabilities
- Recommend mitigation techniques
- Improve the security posture of the web application

Assessment Methodology:

Steps Followed:

1. Reconnaissance and Information Gathering
2. Port Scanning and Service Detection
3. Web Application Vulnerability Scanning
4. Analysis of Identified Vulnerabilities
5. Risk Assessment
6. Recommendation and Mitigation

Tools Used in the Assessment:

Tool Name	Purpose
Kali Linux	Operating System
Nmap	Network scanning and port discovery
OWASP ZAP	Web vulnerability scanning

NMAP SCAN RESULTS



```
kali@kali: ~  
$ nmap demo.testfire.net  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-05 13:33 -0400  
Nmap scan report for demo.testfire.net (65.61.137.117)  
Host is up (0.31s latency).  
Not shown: 996 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
80/tcp    open  http  
443/tcp   open  https  
8080/tcp   open  http-proxy  
8443/tcp   closed https-alt  
  
Nmap done: 1 IP address (1 host up) scanned in 30.16 seconds  
  
kali@kali: ~  
$ nmap -sV demo.testfire.net  
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-05 13:33 -0400  
Nmap scan report for demo.testfire.net (65.61.137.117)  
Host is up (0.39s latency).  
Not shown: 996 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
80/tcp    open  http         Apache Tomcat/Coyote JSP engine 1.1  
443/tcp   open  ssl/http     Apache Tomcat/Coyote JSP engine 1.1  
8080/tcp   open  http         Apache Tomcat/Coyote JSP engine 1.1  
8443/tcp   closed https-alt  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 143.12 seconds  
  
kali@kali: ~  
$
```

This screenshot displays the Nmap scan results used to identify open ports and services running on the target website.

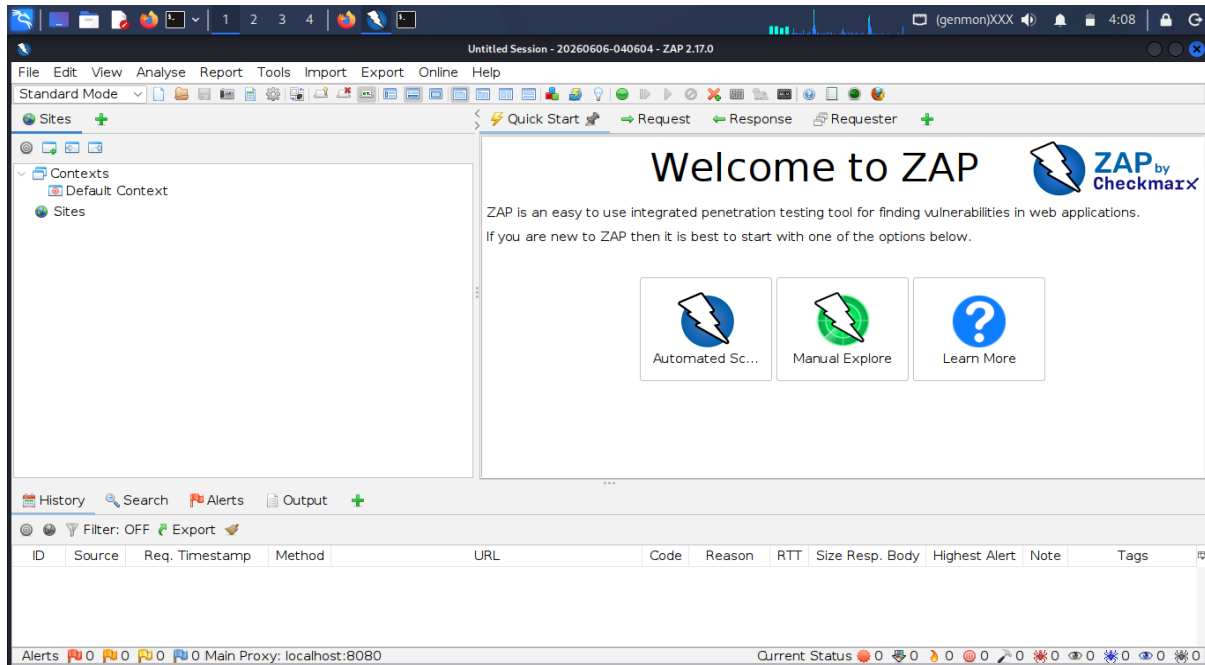
Observed Findings:

- Open HTTP Service
- Open HTTPS Service
- Server Information Disclosure
- Service Version Detection

Recommendations:

- Close unnecessary ports
- Hide service version information
- Restrict public exposure of sensitive services
- Perform regular security audits

OWASP ZAP TOOL



This screenshot shows the OWASP ZAP interface used for passive web application vulnerability scanning. The tool helps identify common security vulnerabilities in websites.

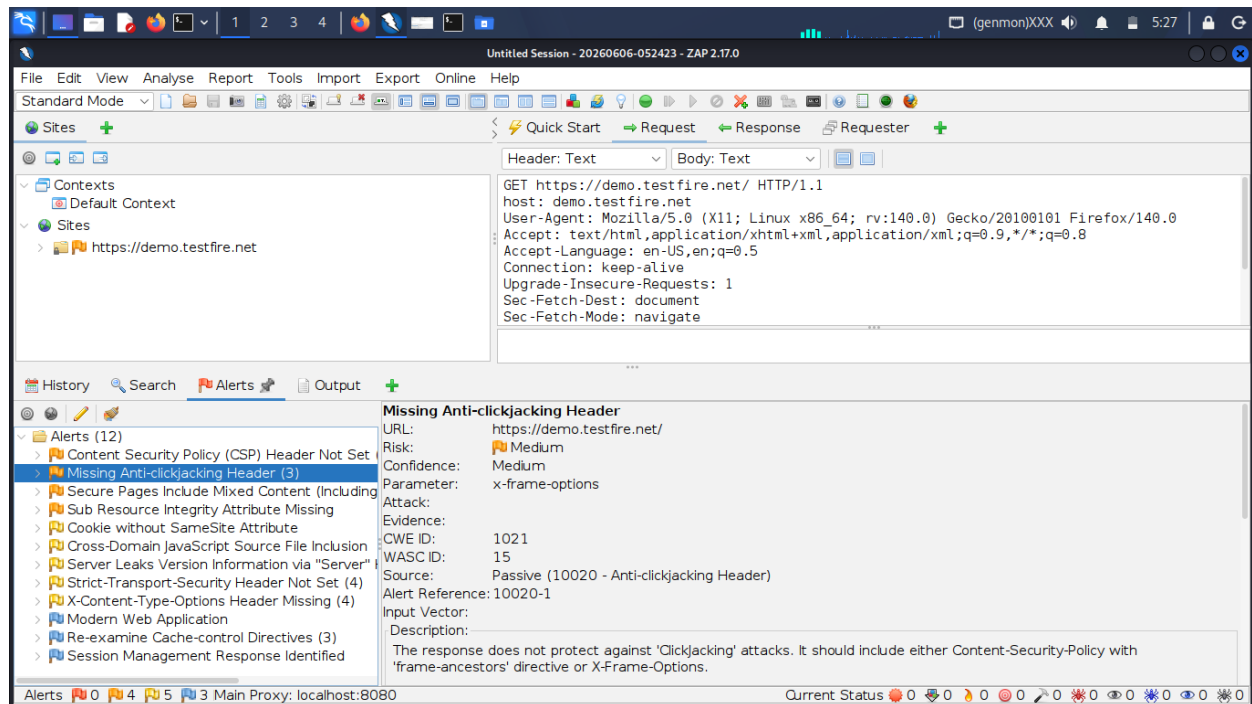
Observed Vulnerabilities:

- Missing Security Headers
- Missing Content Security Policy (CSP)
- Missing Anti-Clickjacking Header
- Missing HSTS Header
- Information Disclosure

Recommendations:

- Configure secure HTTP headers properly
- Enable HTTPS communication
- Implement Content Security Policy (CSP)
- Secure cookie configurations

VULNERABILITY ALERT ANALYSIS



This screenshot displays detailed vulnerability alerts generated by OWASP ZAP during the assessment process.

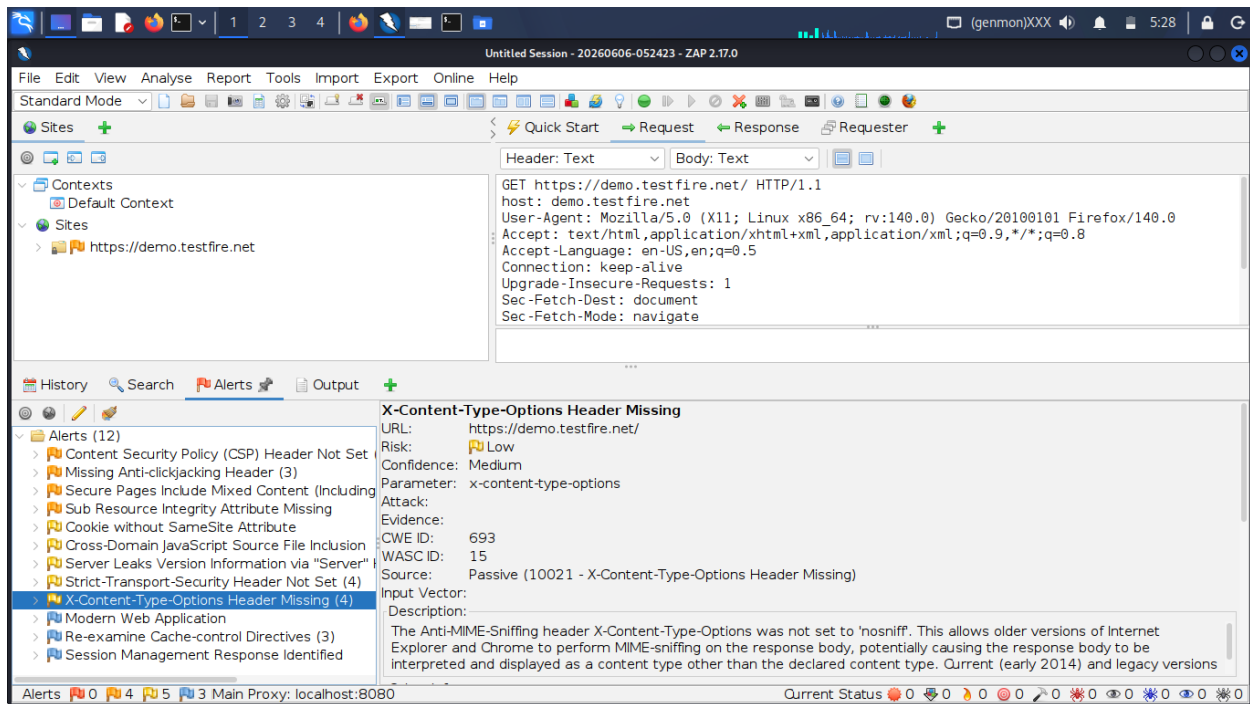
Observed Vulnerabilities:

- Missing Security Headers
- Missing Content Security Policy (CSP)
- Missing Anti-Clickjacking Header
- Information Disclosure
- Missing X-Content-Type-Options Header

Recommendations:

- Enable X-Frame-Options Header
- Enable Content-Security-Policy Header
- Enable X-Content-Type-Options Header
- Configure Strict-Transport-Security Header
- Review web server configurations

Security Alert Information:



This screenshot contains additional security alerts and vulnerability details discovered during the scan

Observed Vulnerabilities:

- Missing Security Headers
- Missing Content Security Policy (CSP)
- Missing Anti-Clickjacking Header
- Information Disclosure
- Missing X-Content-Type-Options Header

Recommendations:

- Configure secure HTTP headers properly
- Use HTTPS communication
- Implement Content Security Policy (CSP)

RISK ASSESSMENT:

Vulnerability Severity

Missing Security Headers Medium

Information Disclosure Medium

Missing CSP Header Medium

Missing HSTS Header Medium

Missing Anti-Clickjacking Header Low

Impact:

- Information Disclosure
- Increased Attack Surface
- Client-Side Security Risks
- Clickjacking Attacks
- Reduced Browser Security Protection

SECURITY RECOMMENDATIONS

- Implement HTTPS across the website
- Configure secure HTTP security headers
- Enable Content Security Policy (CSP)
- Enable HSTS Header
- Validate and sanitize user inputs
- Regularly update software and dependencies
- Monitor logs and suspicious activities
- Perform periodic vulnerability assessments

CONCLUSION:

The vulnerability assessment successfully identified several security weaknesses in the target web application. Using tools such as Kali Linux, Nmap, OWASP ZAP, and Browser Developer Tools helped analyze potential threats and risks.

This assessment provided practical experience in web application security testing, vulnerability analysis, risk assessment, and professional cybersecurity reporting.